Figure 6: **Sampled Images from** Left: Replacing data with data generated by the previous iteration's newly trained VAE yields lower quality and eventually leads to complete mode collapse. Middle: Accumulating data with data generated by the previous iteration's newly trained VAE preserves the quality and diversity of generated data across iterations. Right: Baseline samples after 100 training epochs on the dataset.

the left panel of Figure 6. In contrast, accumulating data at each iteration significantly slows model collapse: the test error increases significantly slower with each additional iteration. While the diversity of generations does go down as compared in the middle and right panel of Fig. 6, it still represents major axes of variation in the dataset, such as gender, but no longer seems to generate other details, along more minor axis of the data manifold, such as glasses and accessories. We discuss further analysis of VAE reconstructions in Appendix D.

Interestingly, unlike language modeling, the test error of accumulating data does increase with the number of iterations (albeit much more slowly than with replacing data). We also note that Martínez et al. (2023a) found slightly contradictory evidence, specifically that a different architecture on a much smaller dataset exhibits fast performance deterioration even with accumulating data. Understanding under what conditions and why these discrepancies exist is an interesting direction we leave for future research.

3 Accumulating Data Avoids Model Collapse in Linear Models

To gain mathematical understanding and intuition, we employ an analytical framework introduced in prior work (Mobahi et al., 2020; Dohmatob et al., 2024a) to understand the difference between data accumulation and data replacement. We will show that it predicts the same types of test error behaviors for these two data-use strategies that were measured empirically. The framework considers a sequence of linear models that are fit to the synthetic data sampled from the linear generative model model based on the previously fit linear models. Within this framework, Dohmatob et al. (2024a) showed that if data are replaced across model-fitting iterations, then the test squared error increases linearly[2] with the number of iterations n. Here, we extend Dohmatob et al. (2024a)'s argument to show that if data instead accumulate across model-fitting iterations, then the test squared error is upper bounded by a relatively small constant, meaning model collapse is avoided[3].

[2]To echo an earlier footnote, an approach 'halfway' between the 'replace' and 'accumulate' approaches would replace the previous dataset with a pure synthetic dataset of size iT at the i-th iteration. Analyzing this goes mostly in parallel, except the $1/i^2$ mentioned in running text now becomes $1/i$ for the 'halfway' approach. Consequently, the MSE scaling becomes $MSE \asymp O(\log(n))$; the 'halfway' approach with pure synthetic data but more of it, again has test error growing unboundedly with iterations. Thanks to Elvis Dohmatob, Yunzhen Feng and Julia Kempe for communicating this observation. See Appendix E for an extended discussion.

[3]In this theoretical section, we identify the term *model collapse* with the situation where test error diverges to infinity (at any rate) as iterations progress. Other authors may employ similar terminology while identifying it with different properties of test error. For example, Alemohammad et al. (2023)

The content in this section relies heavily on the framework and pioneering contributions of Dohmatob et al. (2024a). Our contribution is to study a *different way to use synthetic data in training*, namely accumulate, which seems to better align with certain real-world considerations. We show that our empirical results could have been anticipated on theoretical grounds, by applying the same analysis framework as in Dohmatob et al. (2024a), but instead to this specific training dataset pattern. We use the same framework to analyze some other ways that synthetic data might have be used, such as replace, again the theory aligns with many empirical results.

3.1 Notation and Preliminaries

Original Data Distribution. We adapt notations from Dohmatob et al. (2024a). Define the distribution P_{Σ,w,σ^2} on $\mathbb{R}^d \times \mathbb{R}$ given by $(x, y) \sim P_{\Sigma,w,\sigma^2}$ iff :

$$\begin{aligned} &\text{(Input)} \quad x \sim \mathcal{N}(0, \Sigma), \\ &\text{(Noise)} \quad \epsilon \sim \mathcal{N}(0, \sigma^2), \text{ independent of } x, \\ &\text{(Label)} \quad y = x \cdot w^* + \epsilon. \end{aligned}$$

The positive integer d is the input-dimension, the matrix $\Sigma \in \mathbb{R}^{d\times d}$ is the true covariance structure of the input x, the vector w^* is the true linear relationship used to generate the original data and the scalar σ is the level of label noise. We start at iteration $n = 1$ with T initial independent data points (x_i, y_i) each following P_{Σ,w^*,σ^2}, that is, $y_i = x_i \cdot w^* + \epsilon_i$ for each $i = 1, 2, \cdots, T$. We form the design matrix $X \in \mathbb{R}^{T\times d}$ with $x_1^\top, \cdots, x_T^\top$ as rows. We also form the vectors Y and E with i-th coordinate y_i and ϵ_i respectively. In whatever follows, we will assume that X has full column rank, i.e., $T \geq d$, $X^\top X$ is invertible and the model is underparameterized.

Synthetic Data Generation Process. We generate synthetic data from the following sequence of distributions

$$P_{\Sigma,w^*,\sigma^2} \to P_{\Sigma,\hat{w}_1,\sigma^2} \to \ldots \to P_{\Sigma,\hat{w}_n,\sigma^2},$$

where $n \in \mathbb{N}$ is the number of iterations. The scheme is outlined as follows.

- For $n = 1$:
  - Accumulating Covariates/Features: $\tilde{X}_1 \stackrel{\text{def}}{=} X$
  - Accumulating Targets: $\tilde{Y}_1 \stackrel{\text{def}}{=} \hat{Y}_1 \stackrel{\text{def}}{=} Xw^* + E_1$, where $E_1 \stackrel{\text{def}}{=} E \sim \mathcal{N}(0, \sigma^2 I_T)$
  - Fit linear model: $\hat{w}_1 = \tilde{X}_1^\dagger \tilde{Y}_1$
  - Sample synthetic data for the next iteration: $\hat{Y}_2 \stackrel{\text{def}}{=} X\hat{w}_1 + E_2$, where $E_2 \sim \mathcal{N}(0, \sigma^2 I_T)$
- For $n \geq 2$:
  - Accumulating Covariates/Features: $\tilde{X}_n^\top = [\tilde{X}_{n-1}^\top; X^\top] \in \mathbb{R}^{d\times nT}$
  - Accumulating Targets: $\tilde{Y}_n^\top = [\tilde{Y}_{n-1}^\top; \hat{Y}_n^\top] \in \mathbb{R}^{1\times nT}$
  - Fit linear model: $\hat{w}_n \stackrel{\text{def}}{=} \tilde{X}_n^\dagger \tilde{Y}_n$
  - Sample synthetic data for the next iteration: $\hat{Y}_{n+1} \stackrel{\text{def}}{=} X\hat{w}_n + E_{n+1}$, where $E_{n+1} \sim \mathcal{N}(0, \sigma^2 I_T)$

Here, for a matrix A with full column rank, $A^\dagger = (A^\top A)^{-1} A^\top$ is the Moore-Penrose pseudo-inverse of A. The noise terms $E_1, E_2, \ldots, E_n$ are independent of each other and of the covariates/features. Since X has full column rank, so does $\tilde{X}_n$ for every $n \geq 1$.

use the term MAD to refer to the situation where the distance between the distribution of the original data and that of the subsequent generative models grow farther apart, without necessarily diverging.

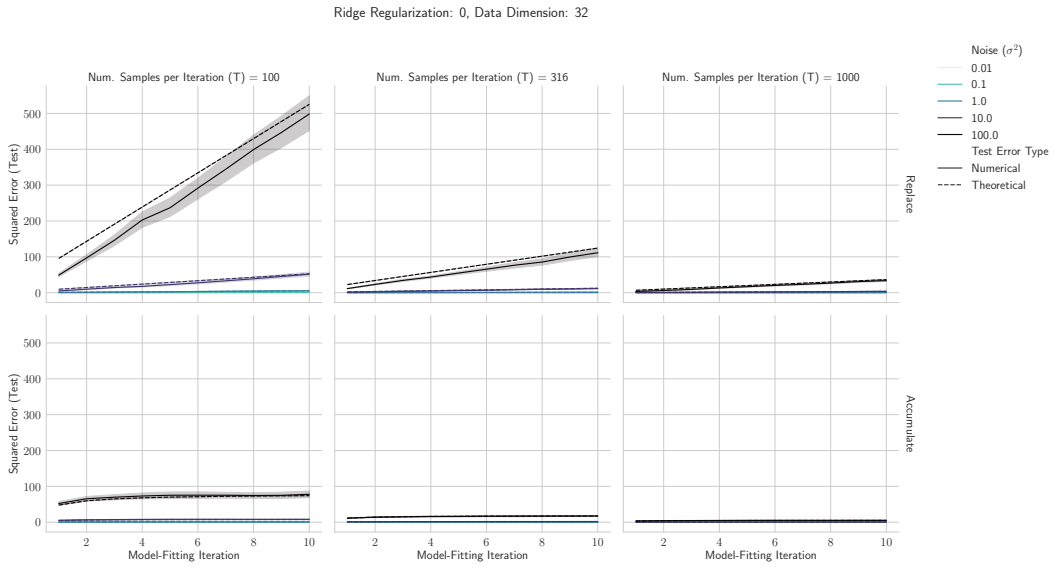

Figure 7: **Accumulating Data Avoids Model Collapse in Linear Regression.** We consider sequences of linear models recurrently fit to generated targets by previous iterations of models. Top: If each linear model is fit to the generated targets of *only* the preceding linear model, i.e., data are replaced, then the test error grows linearly with the number of iterations n. Bottom: If each linear model is instead fit to the generate targets of *all* the preceding linear models, i.e., data accumulate, then the test error has a finite upper bound independent of the number of iterations. This suggests that data accumulation might be a robust solution for mitigating model collapse. For log test error and higher iterations, see Appendix Fig. 16.

Test Error. We are interested in the dynamics of the test error $E_{\text{test}}(\hat{w}_n)$ of this sequence of linear model $\hat{w}_1, \hat{w}_2, \ldots$. Note that evaluation of the model is done on the true distribution P_{Σ,w^*,σ^2}, even though the model is trained on the accumulated synthetic data. For any linear estimator $\hat{w}$ computed from the training data, we measure test error in the standard way:

$$E_{test}(w) \stackrel{\text{def}}{=} \mathbb{E}\left[(x_{test}^T w - y_{test})^2\right] - \sigma^2 = \mathbb{E}[\|w - w^*\|_\Sigma^2] \tag{1}$$

where the expectation is taken over the training data and $(x_{test}, y_{test}) \sim P_{\Sigma,w^*,\sigma^2}$ independent of the training data.

A Note on Extensions to Ridge Regression and Kernel Methods. To reiterate a comment made previously by Dohmatob et al. (2024a), although we present our results in the context of ordinary linear regression in $\mathbb{R}^d$, our analysis can be readily extended to ridge regression and the kernel setting (Caponnetto & De Vito, 2007; Simon et al., 2021; Cui et al., 2021; Wei et al., 2022). We focus here on a simple useful model for studying model collapse.

3.2 Precise Test Error Characterization Under Accumulating Data

Our goal is to establish an analytic formula for the test error of the nth model in the data accumulation setting. We begin by characterizing the relationship between the fitted linear parameters $\hat{w}_n$ and the true parameters w^*. We remind the reader that we assume that X has full column rank, i.e., $X^\top X$ is invertible. Proofs are deferred to App. B.

Theorem 1. *In the data accumulation setting, $\forall n \geq 1$, the fitted linear parameters $\hat{w}_n$ can be expressed as:*

$$\hat{w}_n = w^* + (X^\top X)^{-1} X^\top \left(\sum_{i=1}^{n} \frac{E_i}{i} \right) \tag{2}$$